

Positive global virial work with local Jeans obstructions

Results of the UDCT v7.09 registered GQUMOND example-A test in three ultra-faint dwarfs

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Status. Post-registration numerical report following the v7.09 preregistration, DOI 10.5281/zenodo.23120660. This report does not amend that registration, register a new field law, or claim a new successful UDCT model.

Abstract

The UDCT v7.09 preregistration tests whether the negative density-weighted virial work found in a particular earlier UDCT action also occurs in a specified length-screening GQUMOND example based on Milgrom's construction. The frozen calculation covers Leo IV, Pegasus III and Hydrus I, two stellar profiles, and six screening lengths, with six QUMOND controls. Executing the unchanged companion scripts gives positive normalized virial work in all 36 GQUMOND cases and all six controls. The registered default and smoke grids agree on every virial sign and return Outcome B. An additional, post-registration refined grid confirms this outcome. Positive global work does not establish local equilibrium: the Hydrus I tapered-cusp model has negative conditional isotropic Jeans pressure for screening lengths 1, 3 and 10 pc, and the 10 pc case has outward mean force and negative enclosed effective mass within the registered radial diagnostic interval. The result rules out a universal negative-work interpretation based on the mere presence of a Hessian term within this protocol. It does not identify a unique cause of the older UDCT obstruction, validate MOND generally, or adjudicate bidirectional projection or dynamical back-reaction. The earlier candidate's recorded static-equilibrium problems remain unchanged.

Keywords: GQUMOND; virial work; conditional Jeans equation; Hessian source; preregistration; ultra-faint dwarfs; numerical verification.

1. Question and theory under test

The earlier v7.06 verification note concerns one Hessian-augmented UDCT action family [2]. The present test concerns a different constitutive law, using the example-A structure discussed in Milgrom [3] and frozen in the v7.09 registration [1]. The comparison asks whether the same negative global virial-work obstruction appears after replacing the constitutive law while retaining the registered geometry and numerical protocol. The label “UDCT” names the research series; it does not make the GQUMOND example a new UDCT theory.

The scalar constitutive function is $P = f(u) Q(Z/f)$, with $f(u) = u/(1+u)$, $Z = |\nabla\psi|^2/a_0^2$ and $u = |\nabla\psi|^2/(l_0^2 \psi_{,ij} \psi_{,ij})$. The campaign fixes $Q'(w) = \sqrt{1+w^{-1/2}}$ and $Q(0)=0$, where $w=Z/f$. This is a specified implementation of the example structure, not a test of every GQUMOND function choice. Both the leading gradient source and the action-derived Hessian source are retained. The control sets $f=1$, so the extra Hessian source vanishes.

2. Frozen protocol and registered decision

The objects, profiles, baryonic Galactic field and integration routines come from the unmodified v7.06 dependency script [4]. Stellar mass-to-light ratio is fixed at 2. The density profiles are Plummer and a tapered Hernquist-like cusp, normalized to the same projected half-light radius. The Galaxy supplies a local second-order Taylor field from the visible McMillan disks plus the specified bulge. No dark halo, LMC potential or time evolution is included. The acceleration scale is $a_0 = 1.082401 \times 10^{-10} \text{ m s}^{-2}$.

The registered screening lengths are $l_0 = 0.03, 0.1, 0.3, 1, 3$ and 10 pc . Three objects $\times$ two profiles $\times$ six lengths give **36 GQUMOND cases**; the two profiles for each object give **six QUMOND controls**. These are deterministic parameter-grid calculations, not independent observed systems or random simulation realizations.

The inward-positive angular-mean force defines $V = \int \rho_*(r) r^3 \langle g_r \rangle dr$. Results are divided by the positive stellar Newtonian integral V_N . V is evaluated by radial integration by parts, retaining the boundary contribution. The code also calculates a direct-force integral as a numerical cross-check.

Outcome	Registered global-work condition
A	At least one negative GQUMOND virial case with positive controls.
B	All 36 GQUMOND virial cases are positive.
C	A negative QUMOND control triggers the protocol-validity concern.

The registration allows a virial-sign claim only if default and smoke results agree. No case triggers the registered outward-force warning of more than ten times the stellar Newtonian force in magnitude over $0.1\text{-}10 r_h$; consequently, its conditional $5 r_h$ truncation follow-up is not activated.

3. Global numerical result: Outcome B

Object	Profile	QUMOND V/V_N	GQUMOND min-max
Leo IV	Plummer	14.014605	13.864456 - 14.014603
Leo IV	Tapered cusp	13.896448	13.480055 - 13.896442
Pegasus III	Plummer	19.373652	19.057039 - 19.373649
Pegasus III	Tapered cusp	19.217916	18.461326 - 19.217905
Hydrus I	Plummer	3.994097	3.900006 - 3.994096
Hydrus I	Tapered cusp	3.556883	3.153367 - 3.556879

Table 1. Default-grid normalized virial work. The range spans all six registered screening lengths. Every GQUMOND and control value is positive; the smallest GQUMOND value is 3.153367, for Hydrus I, tapered cusp, $l_0=10 \text{ pc}$.

Outcome B is the registered primary result. The negative global-work obstruction reported for the older UDCT action is not reproduced by this GQUMOND example under the adopted inputs. This is a necessary equilibrium-sign check, not a sufficient demonstration of a physical stellar equilibrium or agreement with observed velocities.

4. Numerical verification and local obstructions

Grid	Radial	Polar $\times$ azimuth	Galactic disk $R \times \varphi \times z$
Smoke	180	16 $\times$ 24	40 $\times$ 24 $\times$ 8
Default	360	24 $\times$ 36	120 $\times$ 64 $\times$ 12
Refined*	720	48 $\times$ 72	240 $\times$ 128 $\times$ 24

Table 2. *Refined grid is an additional post-registration check. It does not replace the registered grids or decision rule. The action-quadrature field inherited in the grid object is unused by the analytic v7.09 constitutive law.

The supplied analytic and geometry self-tests all pass; the dependency hash matches the registered prefix. All 42 virial signs agree on the default, smoke and refined grids. The largest relative default-smoke virial difference is 9.42×10^{-7} ; the largest default-refined difference is 5.57×10^{-8} . The maximum relative disagreement between the refined integration-by-parts virial and its direct-force cross-check is 2.90×10^{-6} . These checks support the reported virial signs; they do not estimate systematic physical-model errors.

The conditional spherical, isotropic Jeans calculation imposes zero radial pressure at the outer boundary: $\rho_* \sigma_r^2(r) = \int_r^{\text{Rout}} \rho_*(s) \langle g_r(s) \rangle ds$. Negative σ_r^2 cannot be interpreted as a physical velocity variance. It signals incompatibility with this reduction, density profile and boundary condition.

Hydrus I cusp l_0 (pc)	Refined min σ_r^2 ($\text{km}^2 \text{s}^{-2}$)	Sampled negative-pressure r/r_h	Sampled outward-force r/r_h
1	-0.113062	0.002 - 0.00470	0.002 - 0.01199
3	-0.435382	0.002 - 0.01837	0.002 - 0.03709
10	-1.571442	0.002 - 0.08475	0.002 - 0.13351

Table 3. Supplementary refined-grid localization of the three negative-pressure cases. Intervals are sampled grid bounds, not root-fitted crossings. The lower mesh limit is $0.002 r_h$; no extrapolation below that limit is asserted. The minima at 1 pc are at this lower boundary.

At $l_0=10$ pc the Hydrus I cusp also has negative effective enclosed mass $M_{\text{eff}}=r^2 \langle g_r \rangle / G$ inside the registered $0.1\text{-}10 r_h$ diagnostic range. The minimum is $-721.24 M_{\text{sun}}$ on the default grid, $-700.34 M_{\text{sun}}$ on smoke, and $-705.25 M_{\text{sun}}$ on the refined grid. The sign is stable, while this sampled minimum varies by approximately 3% between registered grids. It should not be quoted with the precision of the converged global virial.

The 1 and 3 pc outward-force regions are entirely below $0.1 r_h$, so they do not trigger the registered in-range force flag, although the Jeans calculation detects their central negative pressures. All remaining 33 GQUMOND cases and all six controls have no negative pressure on the computed meshes. Negative effective mass or outward force is a static diagnostic here, not a demonstrated prediction of stellar ejection.

5. Complete virial matrix and secondary kinematics

Object / profile	0.03 pc	0.1 pc	0.3 pc	1 pc	3 pc	10 pc
Leo IV / Plummer	14.014603	14.014590	14.014469	14.013099	14.001059	13.864456
Leo IV / cusp	13.896442	13.896381	13.895853	13.890129	13.846229	13.480055
Pegasus III / Plummer	19.373649	19.373620	19.373365	19.370464	19.344974	19.057039
Pegasus III / cusp	19.217905	19.217789	19.216783	19.205843	19.122094	18.461326
Hydrus I / Plummer	3.994096	3.994088	3.994013	3.993156	3.985628	3.900006
Hydrus I / cusp	3.556879	3.556836	3.556455	3.552177	3.516186	3.153367

Table 4. All 36 default-grid GQUMOND values of V/V_N . The profile “cusp” denotes the registered tapered cusp, not an observationally inferred central profile.

Object / profile	QUMOND aperture RMS	GQUMOND formal RMS range
Leo IV / Plummer	1.1237	1.1150 - 1.1237
Leo IV / cusp	1.1904	1.1644 - 1.1904
Pegasus III / Plummer	0.9774	0.9659 - 0.9774
Pegasus III / cusp	1.0335	1.0039 - 1.0335
Hydrus I / Plummer	0.6145	0.5996 - 0.6145
Hydrus I / cusp	0.6485	0.5758 - 0.6485 *

Table 5. Default-grid projected cylindrical-aperture RMS inside one effective radius, in km s^{-1} . *The Hydrus I cusp range includes cases with negative local Jeans pressure; their positive integrated aperture numbers are formal outputs and are not physical dispersion predictions.

Archived dispersion scales in the frozen input are approximately 3.3 km s^{-1} (Leo IV), 5.4 km s^{-1} (Pegasus III) and 2.7 km s^{-1} (Hydrus I). The formal aperture outputs are smaller, but the archived values serve only as scale references here. Matched apertures, observational uncertainties, membership, binaries and non-equilibrium are not modeled. No likelihood, statistical rejection significance, or new observed-data pass count is calculated.

6. Interpretation: what B changes and leaves unchanged

The comparison provides a counterexample, within this protocol and finite parameter grid, to the assertion that adding any action-derived Hessian term must produce negative global virial work in these systems. Different constitutive functions can give different signs. It therefore restricts attempts to generalize the old obstruction to the entire Hessian-dependent theory class.

It does not prove that the integral operation $Q = \int v \, dz$ is inherently defective. Constitutive dependence, scales and derivatives differ between the two candidates; the present comparison does not uniquely isolate which feature causes the older negative result. Nor does Outcome B establish that Milgrom's full theory is observationally correct. The local Hydrus I Jeans obstructions show why “passes the virial-sign gate” is the precise statement.

The previously reported v7.06 problems remain results about that particular candidate under its stated static assumptions. No new UDCT dynamical model is evaluated here, and the historic 0/22 O8d gate is neither rerun nor reopened. A successful research protocol should be distinguished from a successful gravitational theory.

7. Bidirectional projection, limitations and next tests

Bidirectional projection and dynamical back-reaction are not adjudicated by this calculation. Stellar density and the Galactic Taylor field are prescribed; matter and fields are not evolved together. An action-derived Hessian source alone does not demonstrate the proposed bidirectional mechanism. Such a demonstration would require explicit coupled equations, consistent forces and conserved quantities, and evolution from matched initial conditions. The present result supplies neither a confirmation nor an impossibility theorem for that broader proposal.

The physical limitations are the local Galactic Taylor approximation, idealized spherical stellar profiles, fixed mass-to-light ratio, isotropic Jeans reduction and zero outer pressure. Hydrus I's LMC environment is omitted. Refining quadrature resolves numerical concerns within these assumptions; it does not remove them. The next focused tests are central-profile and inner-cutoff sensitivity for the Hydrus I pressure bands, inclusion of a frozen LMC field, and a controlled nonaxisymmetric field/distribution-function calculation. These are proposed follow-ups, not completed results. A replacement UDCT action would require a separate frozen specification and tests.

8. Conclusion

The registered v7.09 primary outcome is B: all 36 GQUMOND cases and six QUMOND controls have positive global virial work, with registered-grid sign agreement and supplementary refinement. Hydrus I's tapered-cusp cases at 1, 3 and 10 pc nevertheless fail the conditional nonnegative-pressure requirement on the computed meshes. The result does not add a direct failure verdict for UDCT's bidirectional or dynamical proposal, and does not rescue the older static candidate. It narrows the scope of the earlier negative-work interpretation and records a useful comparison with explicit remaining obstructions.

Reproducibility and provenance

New numbers in this report come from executing the attached unchanged v7.09 and v7.06 scripts in the Codex environment during this session. Literature and screenshot commentary provide context, not numerical evidence. The original v7.09 files' "not yet run" statements describe their deposit-time status; the preregistration is preserved. Preparation, numerical execution and drafting were assisted by OpenAI Codex. This is an independent research record, not peer review.

Registered execution: `python UDCT_v7_09_Preregistration_GQUMOND_Example_A_Virial_Test_run.py --mode all --convergence`

Supplementary execution: same script with `--mode all --radial 720 --polar 48 --azimuth 72 --disk-radial 240 --disk-azimuth 128 --disk-height 24`. NumPy and SciPy are required.

SHA-256, v7.09 script:

7dc3b908c20bdda90ec91da9f4a687b3
e96e4eee2c817920ced2cc3315762dbf

SHA-256, frozen v7.06 dependency:

a64440e1350ae8bf10eb3188ec34829b
6d586abf275de5cdc841a50ad3478284

References

- [1] W. S. Paik, UDCT v7.09 - Pre-registration: Applying the v7.06 Virial and Jeans Protocol to Milgrom's Published GQUMOND Length-Screening Example, with a QUMOND Control, in Three Ultra-Faint Dwarfs (2026). Zenodo, doi:10.5281/zenodo.23120660.
- [2] W. S. Paik, A sign audit of a Hessian term absent from the standard MOND equation: action-consistent virial work and conditional Jeans tests in three ultra-faint dwarfs, UDCT v7.06, wording revision dated 29 September 2026. doi:10.5281/zenodo.23039754. Historical values are not newly rerun here.
- [3] M. Milgrom, Generalizations of Quasilinear MOND (QUMOND), Physical Review D 108, 084005 (2023). doi:10.1103/PhysRevD.108.084005; arXiv:2305.01589.
- [4] W. S. Paik, v7.09 companion implementation and frozen v7.06 dependency, attached source files. Repository: https://github.com/wspaik5/UDCT_v7.09_Preregistration_GQUMOND_Example_A_Virial_Test. Repository contents were not substituted for the hash-identified attached scripts.